

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	27 June 2026
Project Name	OptiCrop
Maximum Marks	4 Marks

Technical Architecture

The technical architecture of the **OptiCrop – Smart Agricultural Production Optimization Engine** illustrates the overall workflow of the system, including data collection, processing, machine learning-based crop prediction, and result generation. It describes how soil nutrient values (Nitrogen, Phosphorus, Potassium), pH, temperature, humidity, and rainfall data are processed to recommend the most suitable crop. The architecture also includes the technologies used, system components, and the interaction between the user interface, prediction model, and database to ensure accurate and efficient crop recommendations.

OptiCrop – Smart Agricultural Production Optimization Engine

Application / Technology Architecture

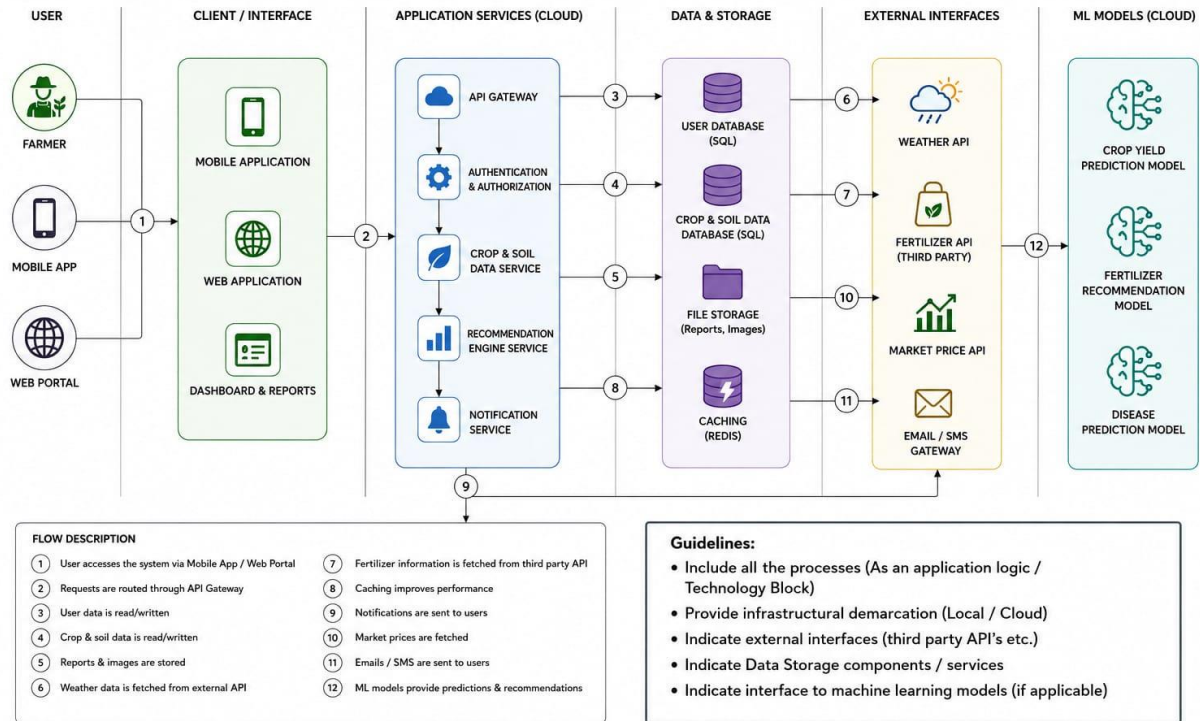


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interface for farmers to enter soil and environmental data and view crop recommendations.	HTML, CSS, JavaScript, Flask
2.	Application Logic-1	Handles user input, validation, and application workflow.	Python
3.	Application Logic-2	Preprocesses soil and environmental data for prediction.	Python (Pandas, NumPy)
4.	Application Logic-3	Predicts the most suitable crop using a machine learning model.	Python (Scikit-learn)

5.	Database	Stores crop details and prediction records (if required).	SQLite
6.	Cloud Database	Not Used	Not Applicable
7.	File Storage	Stores datasets and trained ML model files.	Local File System
8.	External API-1	Not Used	Not Applicable
9.	External API-2	Not Used	Not Applicable
10.	Machine Learning Model	Recommends the most suitable crop based on input parameters.	Random Forest / Decision Tree (Scikit-learn)
11.	Infrastructure (Server / Cloud)	Runs the application on the local computer.	Local System (Flask Development Server)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source frameworks used for developing the web application and machine learning model.	Flask, Scikit-learn, Pandas, NumPy
2.	Security Implementations	Basic input validation and secure handling of user data to prevent invalid inputs.	Python Input Validation, Flask Security Features
3.	Scalable Architecture	Modular architecture with separate layers for user interface, application logic, and machine learning model, allowing future enhancements.	Three-Tier Architecture, Flask
4.	Availability	The application is available on the local system during	Local System (Future: Cloud Deployment)

		execution and can be deployed to the cloud in the future if required.	
5.	Performance	Uses a pre-trained machine learning model for fast crop prediction with efficient data processing.	Scikit-learn, Pandas, NumPy